Exploration of phylogenetic methods applied to superfamily Ostracioidea

Florence Li
BioMath 211 Winter 2026 | Prof. Marc Suchard | UCLA

Introduction

Description of study system

Ostracioidea is composed of 37 species of rigid-bodied fish. The synapomorphy of this superfamily is an armored carapace which gives their bearers their eponymous box shape. This clade is partitioned into two families, Aracanidae and Ostracidae, which evolved body shapes that are distinct from each other. Aracanid species are deeper bodied than the Ostracids, which are characterized by a boxy carapace (Marcroft, 2015). Varying body dimensions lead to different consequences in an organism's fitness. While a boxy Ostracid carapace may sacrifice drag reduction for a defensible body in a shallow, reef environment, an Aracanid's tall and streamlined body shape may trade susceptibility to roll and pitch for a more efficient deep and open water body plan. Morphological studies of Ostracioidea suggest a rigidification of the carapace over time in this clade. Aracanids are thought to possess the more basal body morphology based on this assessment (Tyler, 1980; Klassen, 1995; Parmentier, 2025)).

Applications to evolutionary biology and beyond

Even after two decades of research on boxfish computational fluid dynamics, there is not yet a clear consensus among the literature on how Ostracioidean body shape affects movement efficiency and overall fitness. A major point of debate is whether boxfish body plans exacerbate or diminish drag. Physical models designed after *Ostracion cubicus* showed lower drag than a standard sedan (Chowdury et al., 2019; Kozlov et al., 2015). However, earlier literature suggests that while boxfish fuel economy comparable to body-caudal swimmers, they do so while battling parasitic drag (Gordon et al. 2000). Questions such as these on boxfish fluid dynamics not only investigate the processes underlying the evolution of these unique fish but also inspire biomimetic design. To advance this area of study, it is necessary to infer the phylogenetic relationships between members of this clade.

Past literature & current research

The most recently published phylogeny of Ostraioidea used 2 mitochondrial and 6 nuclear genes from 26 species. The study used both Maximum Likelihood (RaxML) and Bayesian (MrBayes and BEAST) methods to build three trees. The authors found evidence of monophyly in both Aracanid and Ostracids using all three methods. Using two fossil calibrations, they dated the emergence of Ostraioidea to 23 Ma or later (Santini et al. 2013).

In the decade following this study, more sequence information has been collected and shared. This study aims to use newly accessible molecular information to create an updated phylogeny of boxfishes. This study takes advantage of four types of data: morphological, mitochondrial, nuclear, and fossil. Morphological traits were taken from published matrices which were transcribed (Winterbottom & Tyler, 1980; Klassen, 1995; Francesco Santini, unpublished).

Mitogenomes, mitochondrial genes, and nuclear genes were mined from NCBI GenBank. Additional sequences and fossil information were downloaded from Fish Tree of Life (Rabosky et al. 2018). This study tested a variety of tree-building methods including phangorn, IQ-TREE + ASTRAL, MrBayes, and BEAST2. I expected this richer dataset to group Aracanids and Ostracids separately, maintaining their monophyly. In addition, I anticipated differences in sister groupings within these families between the three methods as well as between these trees and those published in Santini et al. 2013.

Methods

Data procurement

A morphological character matrix of Ostracioidea based on Winterbottom & Tyler, 1980 and Klassen 1995 and transcribed by Francesco Santini was used for the maximum parsimony and Bayesian methods. NCBI produced 12 complete and 1 partially complete mitochondrial genomes and additional gene sequences. Fish Tree of Life supplied supplemental sequence information and three fossil calibrations (Table 1). The resulting data assembly showed adequate coverage for 26 out of the 37 known species of boxfish (Fig. 1). Triodon macropterus was used as an outgroup due to its established position in the tree of life and abundance of molecular data.

My goal was to procure as much data as possible across as many species as possible. From those data, I used sequence length as a proxy for using the best quality sequences. Mitogenome sequences were prioritized and the rest of the gene matrix was filled in with the best quality gene barcodes from NCBI and Fish Tree of Life.

Sequence alignment

The 13 mitogenomes were aligned pairwise using LASTZ against a reference genome, NC_009865.1, *Ostracion immaculatus.* Then, gene coordinates from the reference genome metadata were used to extract barcoding mitochondrial loci. MAFFT was then used to align the extracted loci and gene sequences from other sources to each other. I concatenated 15 mitochondrial genes, 12 nuclear genes, and 120 morphological characters into a supermatrix. Six taxa shown in Fig. 1 with no, or minimal data were excluded.

Preliminary methods tests

I built four preliminary trees using a variety of methods using phangorn, IQ-TREE, and MrBayes. The phangorn tree used exclusively the morphological data matrix to create a maximum parsimony tree rooted on *Kentrocapros aculeatus*. I used Parsimony Ratchet (function prachet()) which uses the Fitch algorithm with additional perturbations (adding random weights) to various sites as a method of escaping local peaks (Nixon, 1999). It then compares the perturbed to the unperturbed tree and chooses the lowest score.

I then generated two maximum likelihood trees using IQ-TREE using two datasets: the 13 mitogenomes and the nuclear genes downloaded from Fish Tree of Life. For the mitogenome tree, IQ-TREE used a continuous-time Markov chain (CTMC), meaning that it didn't discretize time and only considered the current character state, not states in the past. It used a general

time reversible +F+G4 model (GTR) meaning that the rate of substitution from one character to another is the same as the probability of its reverse. F refers to that the model considered the existing frequencies of basepairs in the sequences. G4 refers to gamma distribution meaning that substitution rates are sorted into four different bins. For this method, mitogenomes were treated as one single independent, non-recombining locus.

For the nuclear likelihood tree, IQ-TREE used ModelFinder, which used different mathematical models for the independent nuclear loci. Table 2 shows which types of models were used for which genes. Group 4, Kimura, is the simplest model assuming equal frequencies with the rate of transition not equalling the rate of transversion (Kimura, 1980). Group 3, Tamura Nei (TN), further distinguishes between the rate of transition between purines and that between pyrimidines (Tamura & Nei, 1993). Groups 1 and 2 (TIM2 &TPM2u) use even more specific substitution rates and distinctions.

Lastly, I test-ran MrBayes with the following assumptions: GTR+G which differs from the IQ-TREE mitogenome analysis (GTR+F) in that the basepair frequency is assumed rather than accounted for. It used the Mk model (Lewis, 2001) for the morphological data, which is a CTMC assuming equal transition rates between all character states. It ran for 2,000,000 generations with four chains (one cold + three heated) and a sampling frequency of 500 generations per sample. There were two runs and a burnin of 12.5% (discarding the first 500 samples).

Secondary methods tests
For the second round of tree generations, I used the procured dataset described in *Data procurement* to create trees with IQ-TREE and BEAST2. In these runs, I treated mitochondrial genes as independentI ran IQ-TREE using only the molecular matrix using GTR+G with 1000 bootstraps. Then I used ASTRAL which uses coalescent theory to analyze the 27 gene trees for the most likely topology.

For the Bayesian tree, I used BEAUti to create a .xml file with the following specifications. All molecular partitions used GTR+G4 and the morphological partition used Mk. All molecular partitions used a Relaxed Clock Log Normal and the morphological partition used a strict clock model. I used a Birth-Death model as a prior as well as three fossil calibrations as listed in Table 1. BEAST used a chain length of 20,000,000 with a sampling frequency of once every 2000 generations, 4 independent chains, and a burnin of 25%. Then I combined the 4 trees with LogCombiner and made a maximum clade credibility tree with TreeAnnotater with node height set at mean height.

Clade	**Fossil**	**Minimum age (Ma)**	**Included taxa**
Total Group Triodontidae	*Ctenoplectus williamsi*	53.7 Ma	All
Total Group Ostraciidae	*Eolactoria sorbinii*	49 Ma	All except *Triodon macropterus*

Crown Group Ostraciidae	*Oligolactoria bubiki*	30.28 Ma	Only Ostraciidae

Table 1. Fossil calibrations used in BEAST2 method taken from FishTreeofLife.org

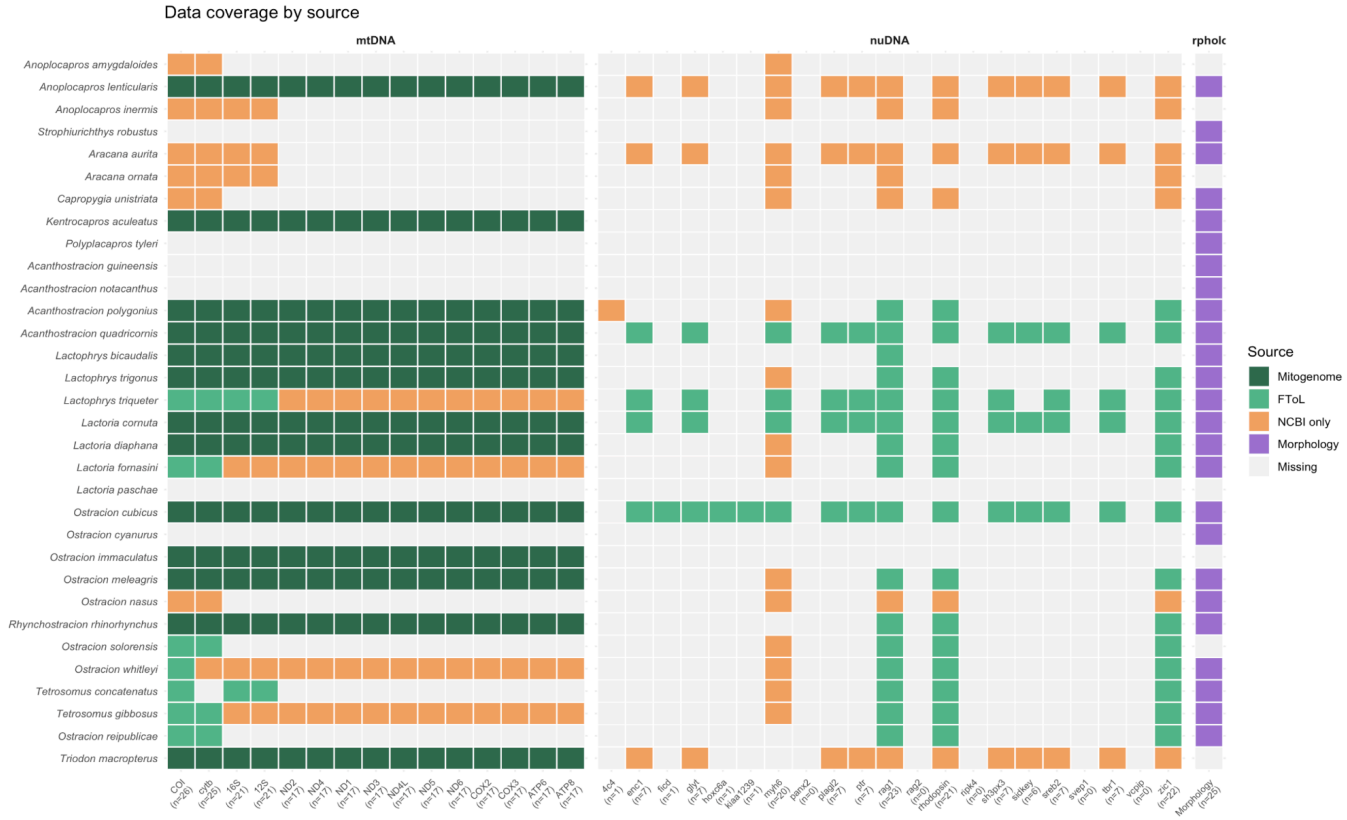


Figure 1. Data coverage across loci and taxa color coded based on source. Mitogenome refers to mitochondrial genomes taken from NCBI. NCBI only refers to gene sequences from NCBI.

Model Group	Substitution Model	Frequencies	Rate Heterogeneity	Genes / Loci Included
Group 1	**TIM2** (Transitional Model 2)	**+F** (Empirical)	**+I** (Invariable sites)	12s, 16s, hoxc6a
Group 2	**TPM2u** (Transitional PM 2 unequal)	**+F** (Empirical)	**+I+G4** (Invariable + Gamma)	4c4, coi, cytb, ripk4
Group 3	**TN** (Tamura-Nei)	**+F** (Empirical)	**+G4** (Gamma)	enc1, ficd, kiaa1239, myh6, nd2, nd4, plagl2, ptr, rag1, rag2, rhodopsin, sh3px3, sidkey, sreb2, svep1, tbr1, vcpip, zic1
Group 4	**K2P** (Kimura 2-Parameter)	*Equal* (Built-in)	**+I** (Invariable sites)	glyt, panx2

Table 2. Nuclear loci grouped by which model type chosen by ModelFinder for IQ-TREE

Results

Preliminary results

All four preliminary trees grouped Ostracids in a monophyletic clade. They also all show *Lactrophrys* as sister to *Acanthostracion*. The nuclear IQ-TREE tree shows Lactoria as sister to Ostracion with Tetrosomus sister to the former two while the MrBayes tree shows Tetrosomus as nested within Lactoria. The Bayesian tree had ESS for morpho shape = 65 (<200) and for partition rate multipliers = 45 (<200).

Secondary results

Figures 3 and 4 show Aracanid and Ostracid monophyly and monophyly of most genuses (*Ostracion reipublicae* is synonymous with *Tetrosomus reipublicae)*. IQ-TREE shows *Kentrocapros* as sister to *Capropygia* while BEAST shows it as sister to the rest of the Aracanids. The two trees disagree about which two of the three *Lactophrys* taxa are sister to each other. Both trees show *Tetrosomus* as nested within *Lactoria*. They also disagree about the specific topology within *Ostracion*. This tree dates the separation of the two families at the beginning of the Oligocene. The ESS values for this tree were chain 1: 114, chain 2: 123, chain 3: 121, chain 4: 66 all below 200.

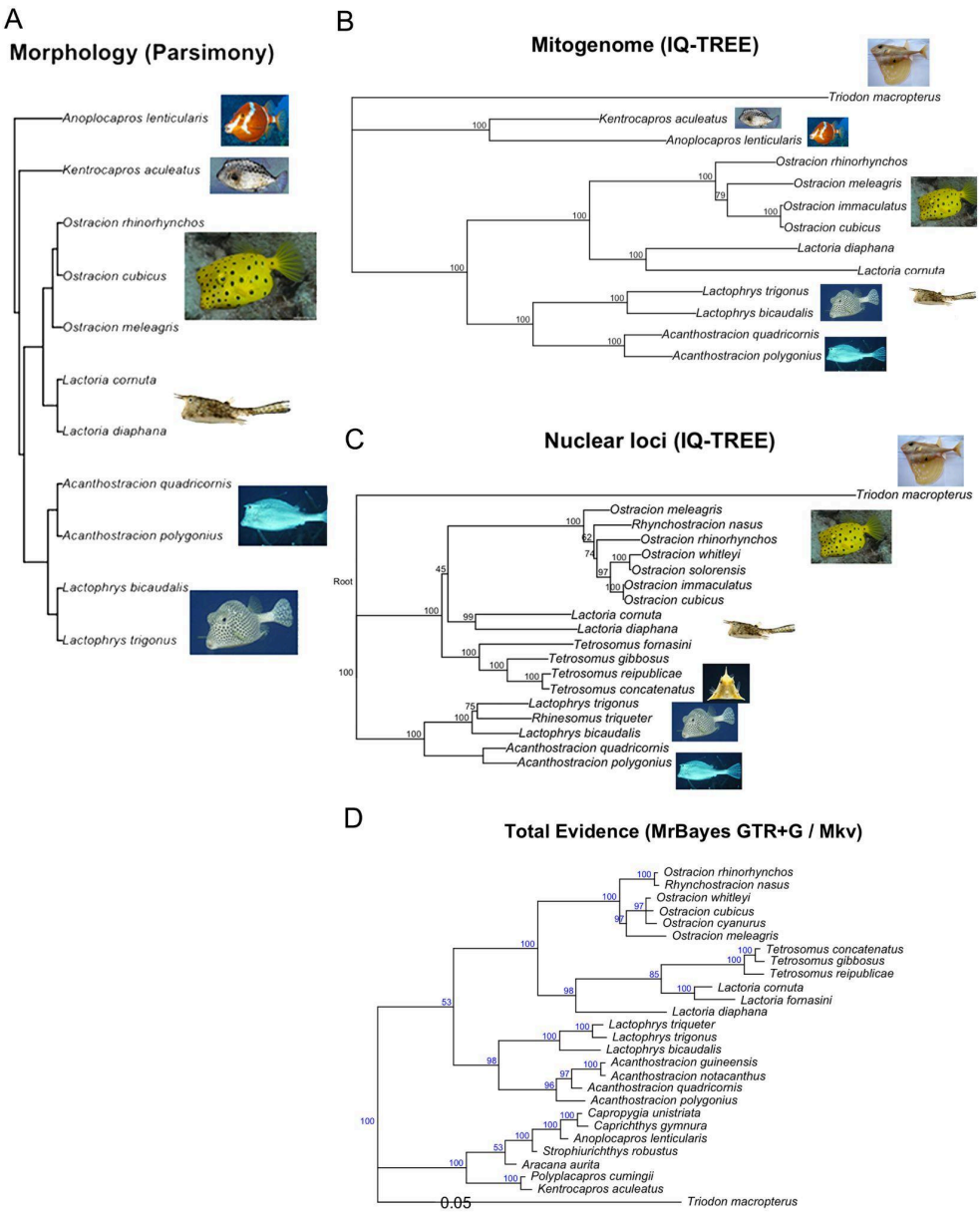


Figure 2. Preliminary trees A. parsimony tree made with phangorn using the morphological matrix alone B. maximum likelihood tree made with IQ-TREE using mitochondrial genomes (no recombination) C. maximum likelihood tree made with IQ-TREE using nuclear loci (recombination) D. Bayesian tree made using MrBayes using morphological matrix, mitogenomes, and nuclear loci.

IQ-TREEASTRAL coalescent species tree

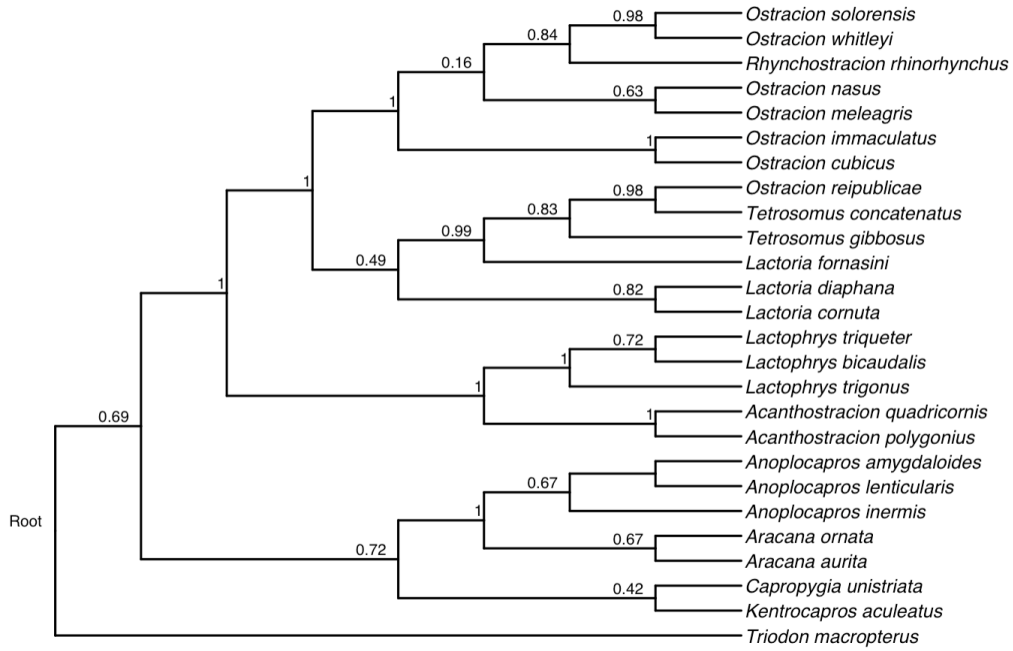


Figure 3. Maximum likelihood tree made with IQ-TREE and ASTRAL using most complete molecular dataset

BEAST total evidence timetree

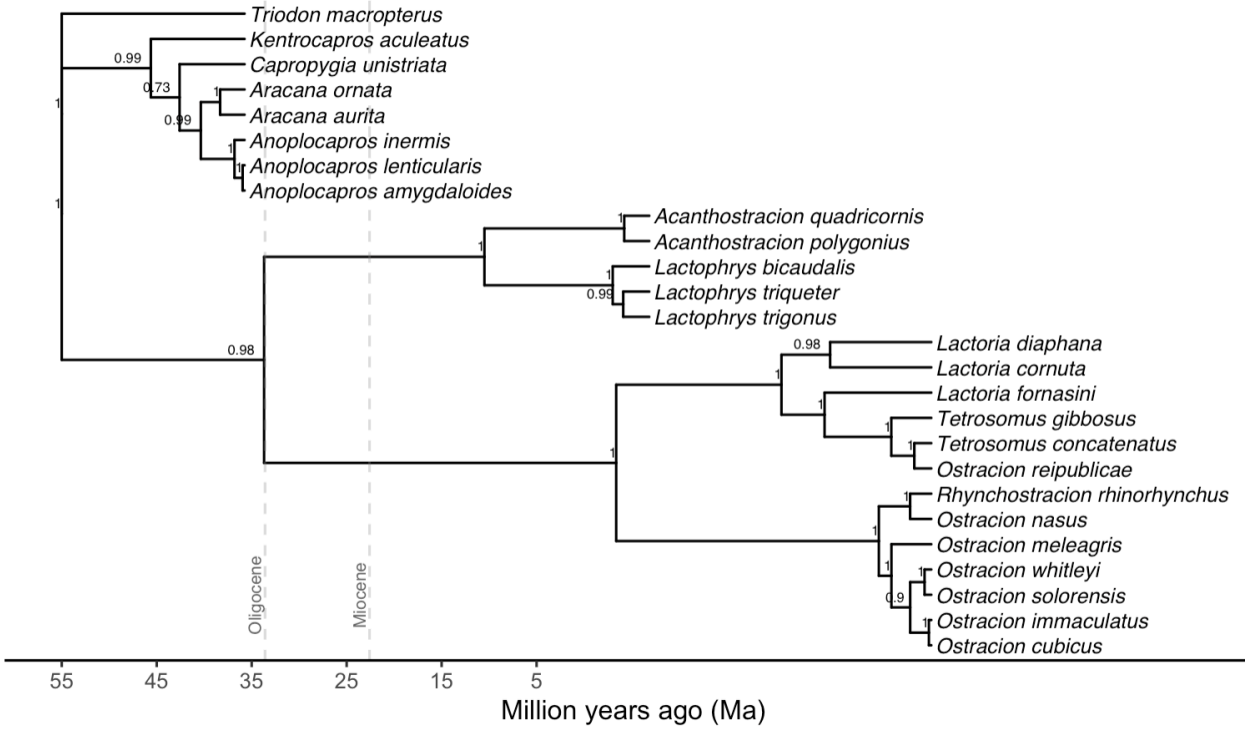


Figure 4. Bayesian tree made with BEAST2 using most complete morphological and molecular dataset

Discussion

These results generally agree with the previous phylogenies published by Santini et al. 2013. Both of the Bayesian trees were limited by computation time and did not fully converge. Despite this, they still present a topology that is mostly similar to that of the likelihood trees. These results matched my expectations. The time calibration on this BEAST tree compared to that form the Santini et al. 2013 paper is vastly different with this tree suggesting much younger clades. Due to the low effective sample size of this BEAST tree, these calibrations may be unreliable.

Further work

The next steps are to go back through the literature to look at synonymizations, to comb through other databases for additional sequence information for missing taxa, and to remake a tree using BEAST2 with a longer run time to allow for convergence.

Applications

Resolving the boxfish phylogeny is an important step to studying their evolution with possible applications for biomimetic design. With a robust phylogeny, we can better understand how selection might have acted on boxfish body shapes. Lessons in boxfish fluid dynamics can then be applied to robotics and vehicular designs. Boxfish are unique in that they represent a natural system that faces the same design challenges as an artificial autonomous vehicle: how do you maneuver a boxy, rigid-bodied structure through fluid as efficiently as possible?

References

Kimura, M. (1980). A simple method for estimating evolutionary rates of base substitutions through comparative studies of nucleotide sequences. *Journal of molecular evolution*, *16*(2), 111-120.

Klassen, G. J. (1995). Phylogeny and biogeography of the Ostraciinae (Tetraodontiformes:Ostraciidae). *Bulletin of Marine Science*, *57*(2), 393-441.

Lewis, P. O. (2001). A likelihood approach to estimating phylogeny from discrete morphological character data. *Systematic biology*, *50*(6), 913-925.

Marcroft, T. A. (2015). *Evolution of the boxfish carapace: functional consequences of shape*. University of California, Los Angeles. https://www.proquest.com/openview/b61421f49182abf1207655df2776e41b/1?pq-origsite=gscholar&cbl=18750

Nixon, K. C. (1999). The parsimony ratchet, a new method for rapid parsimony analysis. *Cladistics*, *15*(4), 407-414.

Parmentier, E., Eche, L., Greeven, C., Banse, M., Bertucci, F., Thiry, M., ... & Lecchini, D. (2025). Morphological innovations and evolutionary transitions in boxfish acoustic communication. *Biological Journal of the Linnean Society*, *146*(2), blaf079.

Rabosky, D. L., Chang, J., Cowman, P. F., Sallan, L., Friedman, M., Kaschner, K., ... & Alfaro, M.E. (2018). An inverse latitudinal gradient in speciation rate for marine fishes. *Nature*, *559*(7714), 392-395.

Santini, F., Sorenson, L., Marcroft, T., Dornburg, A., & Alfaro, M. E. (2013). A multilocus molecular phylogeny of boxfishes (Aracanidae, Ostraciidae; Tetraodontiformes). *Molecular phylogenetics and evolution*, *66*(1), 153-160.

Tamura, K., & Nei, M. (1993). Estimation of the number of nucleotide substitutions in the control region of mitochondrial DNA in humans and chimpanzees. *Molecular biology and evolution*, *10*(3), 512-526.

Winterbottom, R., & Tyler, J. C. (1983). Phylogenetic relationships of aracanin genera of boxfishes (Ostraciidae: Tetraodontiformes). *Copeia*, 902-917.